

hyper-thermochem: JAX-compatible and differentiable local kernels for hypersonic nonequilibrium CFD

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Local Closure Layer

Transitional hypersonic flows require closures that remain usable under strong thermochemical nonequilibrium, stiff source terms, and changing collisional regimes. `hyper-thermochem` is a reusable local property and source-term library for that setting: thermodynamics, chemistry, relaxation, transport, and equilibrium are evaluated pointwise, while the CFD solver keeps ownership of mesh, gradients, fluxes, time integration, and boundary conditions.

CFD-Facing Interface

The public interface is intentionally local:

```
context → state → requested closures
```

A reusable context is built once per mixture/model choice; a state is built per cell or quadrature point from ϱ , φ_α , and modal temperatures ϑ_t .

Typical requests include:

- thermodynamics: p , ψ , ε_t , mode chemical potentials,
- chemistry: species source terms τ_α ,
- relaxation: modal source terms π_t ,
- transport: viscosity, conductivity, and diffusion operators,
- equilibrium: constraint-consistent local compositions via `equilibrate.py`.

The same state maps directly to conservative variables used by a solver,

$$\varrho e = \varrho \left(\sum_{i \in \{t, r, v, e\}} \varepsilon_t + \frac{1}{2} |\mathbf{v}|^2 \right),$$

with modal energy densities stored in the canonical (t, r, v, e) basis.

Mixtures, Mode Models, and Validation

Current bundled mixtures are `air5`, `air11`, and `mars19`.

Mode model	1T	2T	3T	4T
temp_map	[0,0,0,0]	[0,0,1,1]	[0,0,1,2]	[0,1,2,3]

Default transport closures include Chapman–Enskog viscosity and thermal conductivity with Stefan–Maxwell diffusion.

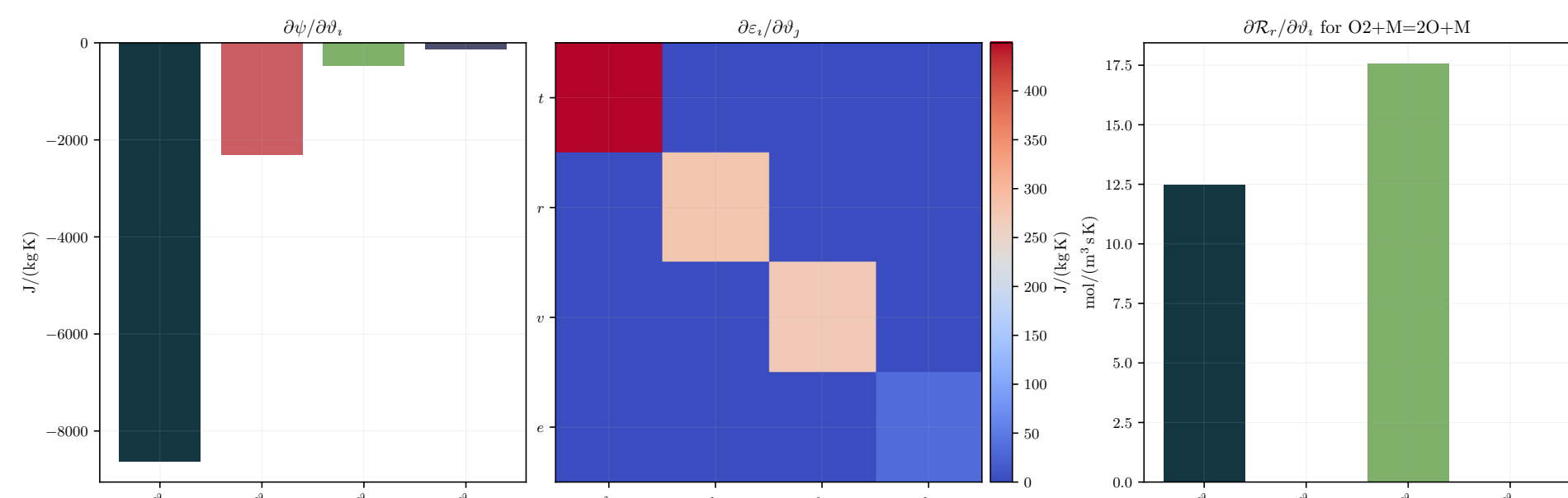


Figure 1. Selected thermodynamic and chemistry sensitivities for the baseline air5 parcel.

Thermochemistry and Conservation Checks

In the common-translational-temperature specialization, the rigid-rotor harmonic-oscillator free-energy kernels recover

$$\varepsilon_t = \frac{3}{2} \bar{R} \vartheta_t, \quad p_t = \varrho \bar{R} \vartheta_t, \quad \bar{R} = \sum_{\alpha \in \mathcal{C}} \varphi_\alpha R_\alpha.$$

Species production τ_α and modal exchange π_t use fixed constituent and (t, r, v, e) orderings and satisfy

$$\sum_{\alpha \in \mathcal{C}} \tau_\alpha = 0, \quad \sum_{t \in \mathcal{D}} \pi_t = 0,$$

where $\mathcal{C}$ and $\mathcal{D}$ are the constituent and mode sets. The CFD solver retains ownership of spatial fluxes, stress power, and time integration.

Selected Thermodynamics and Chemistry Maps

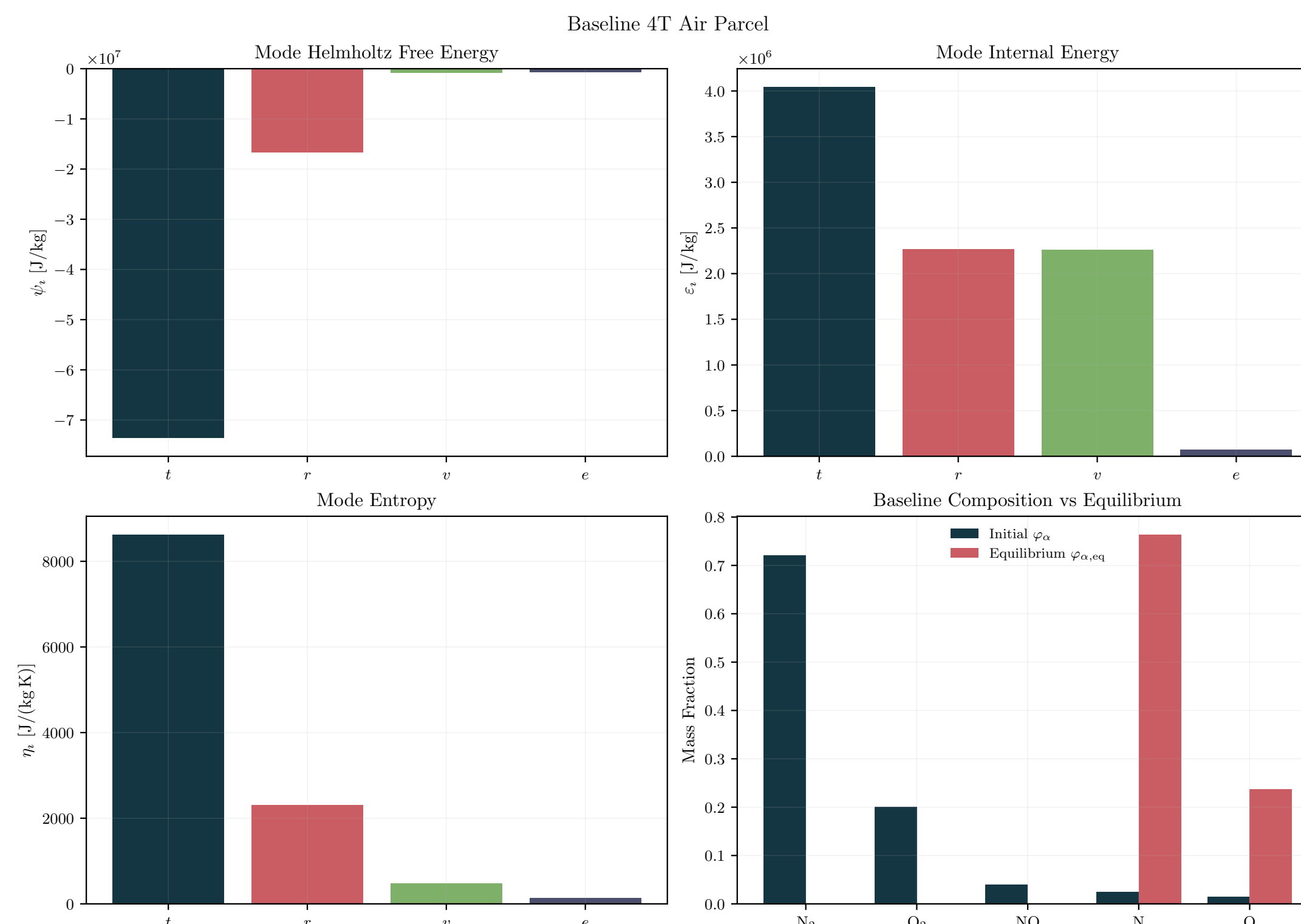


Figure 2. Baseline mode decomposition and initial-to-equilibrium composition shift for air5.

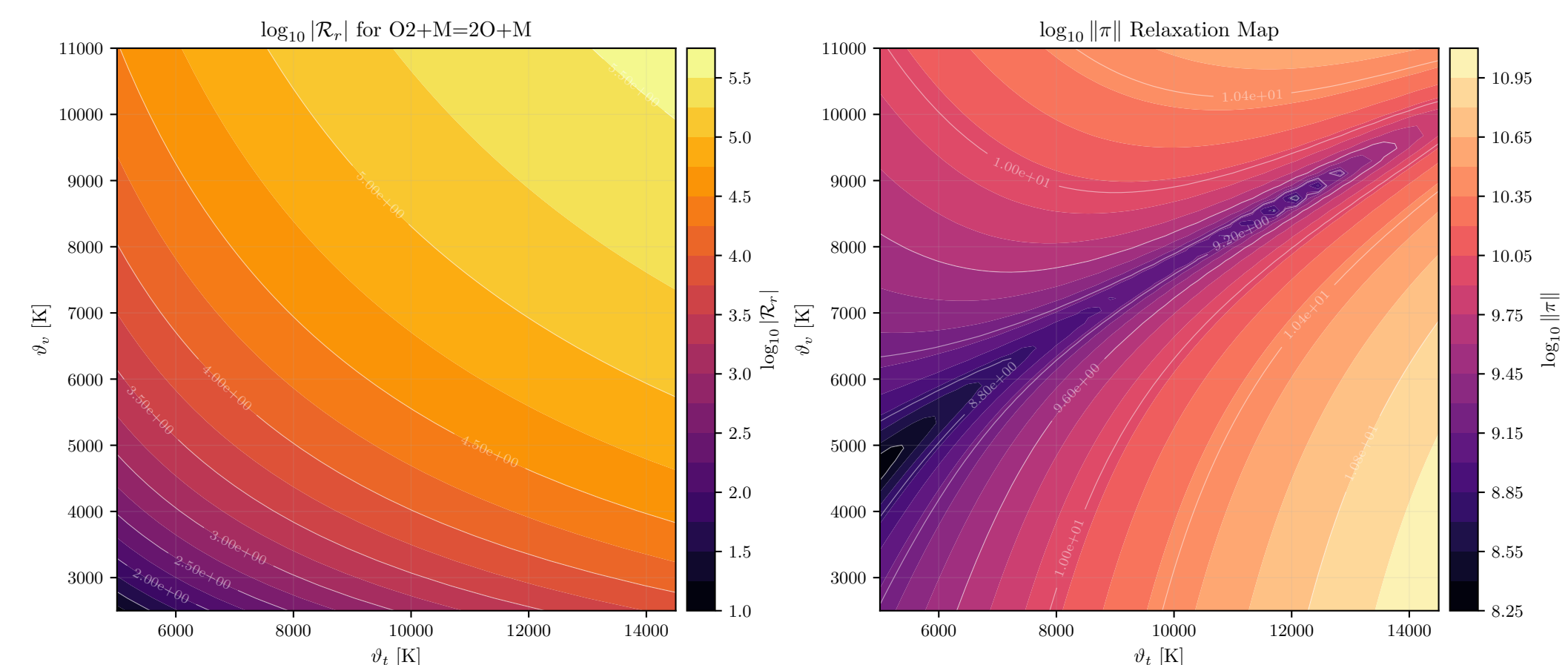


Figure 3. Finite-rate chemistry response on the same nonequilibrium plane.

Charged-Mixture Interface Boundary

Charged support is deliberately local. Given a scalar electric potential supplied by the host solver, the thermodynamic kernel returns relative mode-wise electrochemical potentials of the form

$$\tilde{\mu}_{\alpha t} := \mu_{\alpha t} + \mathbb{I}_{\{t \equiv m\}} (q_\alpha - q_\sigma) \phi,$$

where σ is the reference constituent. The library also supplies charge density, static permittivity, electric conductivity, and native charged Ramshaw or Exact/Stefan–Maxwell coefficient matrices.

These charged matrices are transport-coefficient outputs, not a validated thermodynamic mobility. They must therefore be used with their native driving-force definitions rather than contracted directly with gradients of the electrochemical potentials above.

The host solver owns spatial gradients, electric-field or quasineutral closure, flux assembly, electrostatic field energy and stress, boundary conditions, and time integration. The library does not solve a field equation or add Maxwell stress to the material pressure.

Why This Separation Matters for CFD

Separating spatial numerics from stiff local physics lets a solver:

- request only the closures needed for a given model,
- detect conservation or consistency failures close to the source term,
- reuse the same local kernels across neutral air, ionising air, and Mars-entry cases.

Transport and Equilibrium Maps

Beyond scalar thermodynamic properties, the adapter returns field-shaped transport objects needed by high-temperature CFD workflows: heavy-particle conductivity, modal conductivities, multicomponent diffusion matrices, and charged-mixture diffusion variants, together with local equilibrium reconstruction.

The current library exposes neutral Exact mobility and native charged Ramshaw and Exact/Stefan–Maxwell coefficient matrices. The latter are not charged thermodynamic mobilities; the CFD solver constructs their native spatial driving forces and the corresponding fluxes.

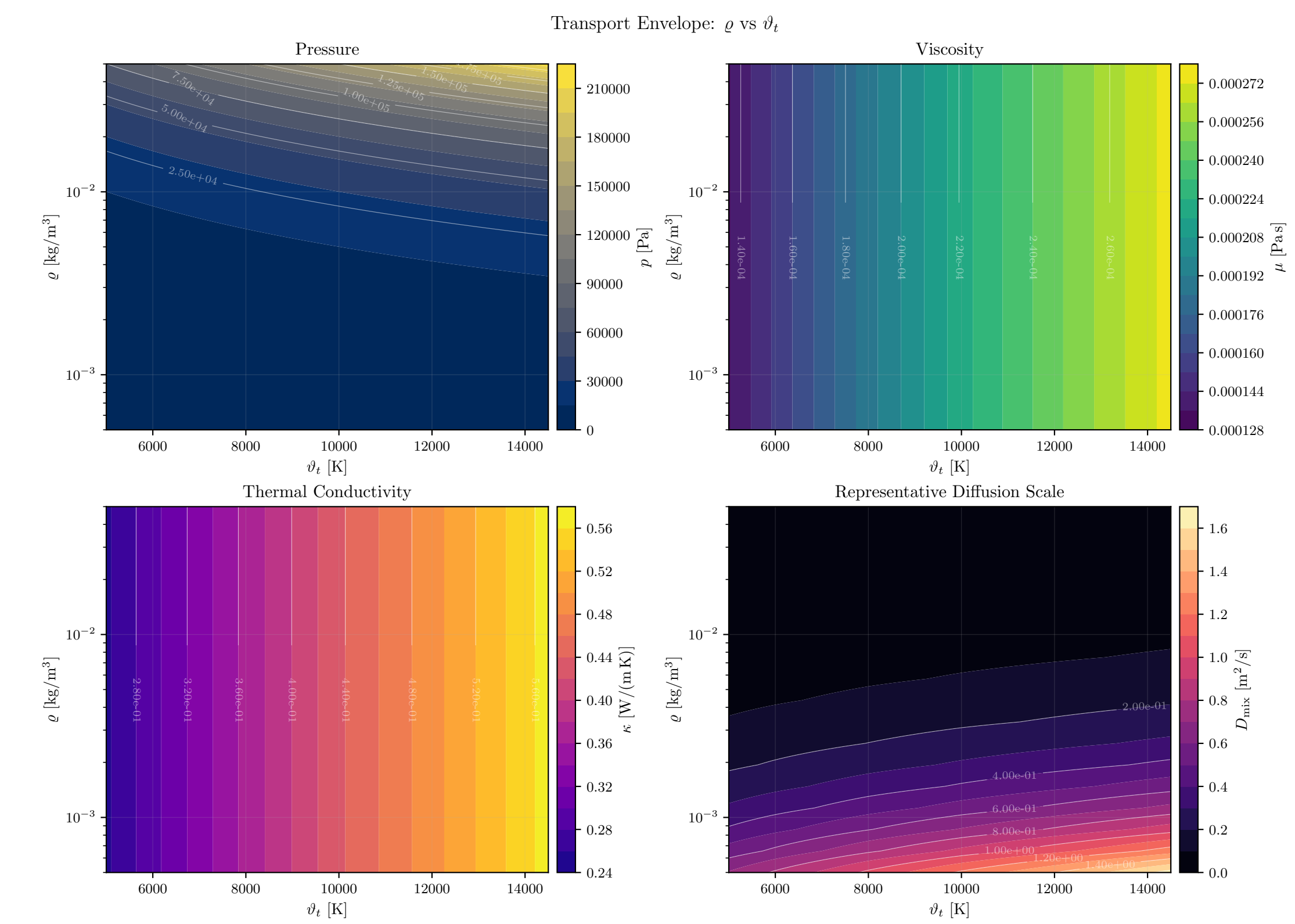


Figure 4. Air5 transport variation with density and translational temperature.

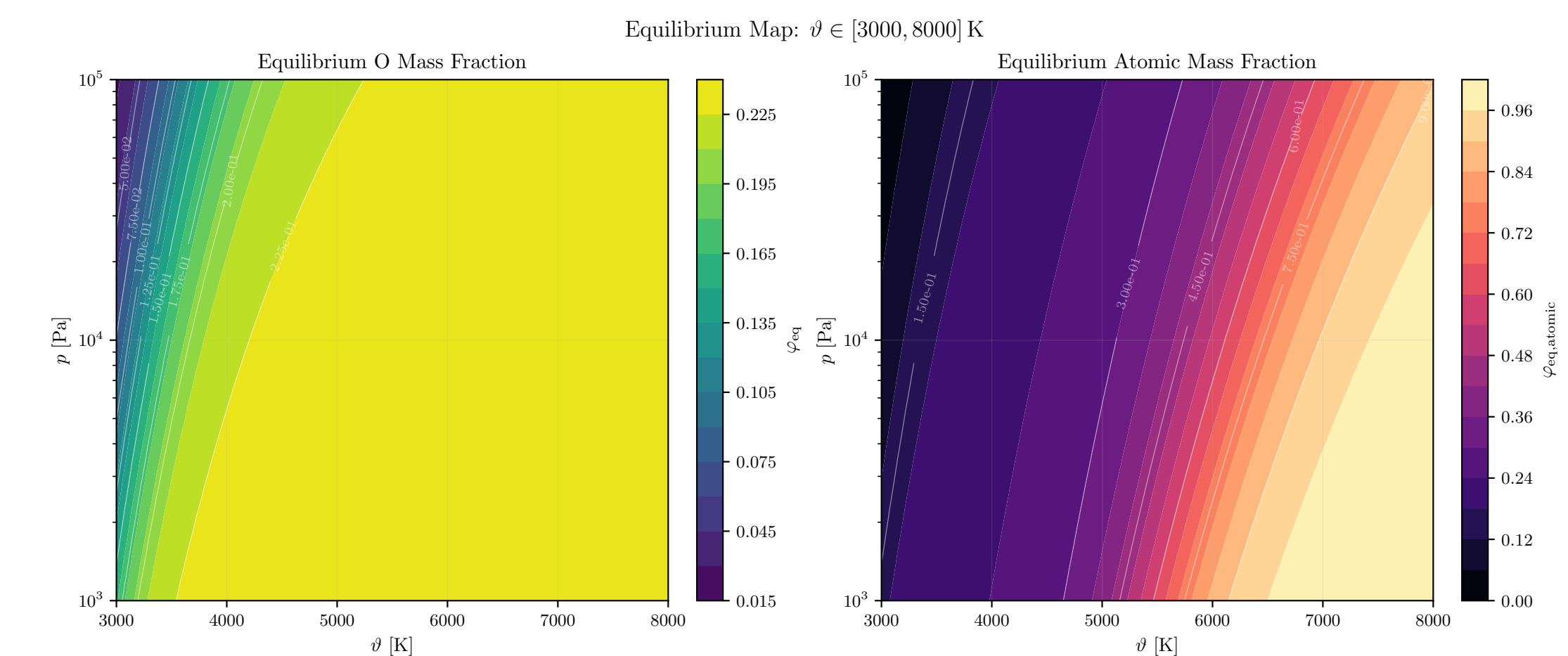


Figure 5. Focused equilibrium map for oxygen and atomic fractions over temperature and pressure.

Implementation and Verification

The library provides:

- bundled inputs and public adapters for `air5`, `air11`, and `mars19`,
- smoke tests, regression tests, and Mutation++ comparison scripts,
- consistent interfaces for thermodynamics, chemistry, transport, relaxation, and equilibrium.